



Project Title: *"Market Basket Analysis"*

Group Members: Ayesha Ali (23k-7317)

Course Name: Marketing Analytics

Instructors Name: Maam Asiya

Date of Submission: 6th Dec 2024

Contents

- 1. Introduction to Market Basket Analysis** 3
 - What is Market Basket Analysis? 3
 - Why is Market Basket Analysis Important? 3
 - Real-World Applications of Market Basket Analysis 3
- 2. Purpose of the Project** 4
 - Why Perform Market Basket Analysis on This Dataset? 4
 - Understanding Customer Behavior: 4
 - Optimizing Business Operations:..... 4
 - Addressing Data Science Objectives: 4
 - Business Questions and Challenges Addressed 4
- 3. Dataset Overview** 6
 - Describe the Dataset Structure 6
 - Highlight Key Statistics 7
- 4. Analysis**..... 7
 - Sales Distribution of Items 7
 - Top 10 Most Popular Items Sold 8
 - Customer Behavior Analysis 9
 - Apriori algorithm10
- 5. Key Insights from the Market Basket Analysis:**.....11
- 6. Recommendations:**12
- 7. Business Benefits:**13
- 8. Conclusion:**13

1. Introduction to Market Basket Analysis

What is Market Basket Analysis?

Market Basket Analysis is a technique used in data science to discover patterns and relationships within large sets of transactional data, such as sales or purchase records. It focuses on identifying items that are frequently bought together. For example, if customers who buy bread often purchase butter as well, this insight can help businesses optimize their operations.

Why is Market Basket Analysis Important?

This analysis is especially valuable for businesses in retail and e-commerce because it allows them to:

1. **Understand Customer Behavior:** It reveals customer purchasing habits, such as common combinations of products they buy.
2. **Optimize Store Layouts:** Retailers can place related products closer together to increase sales.
3. **Improve Marketing and Promotions:** Businesses can create product bundles or targeted promotions for frequently purchased items.
4. **Enhance Cross-Selling Opportunities:** Recommending related products encourages customers to buy more.

Real-World Applications of Market Basket Analysis

Market Basket Analysis is used extensively by leading companies to boost profits and customer satisfaction. Below are some examples:

1. **Amazon:** When you shop online at Amazon, you may notice a section that says "Frequently Bought Together." These recommendations are generated using Market Basket Analysis to suggest items often purchased together. For instance, if a customer buys a laptop, they might be shown accessories like a laptop bag or mouse.
2. **Walmart:** Walmart uses this technique to optimize inventory and product placement. A famous example is when Walmart found that people buying diapers often purchased beer. They placed these items closer together in stores, leading to increased sales.
3. **Netflix:** In the entertainment industry, Netflix uses a similar approach to recommend movies or shows based on viewing habits, leveraging customer preference patterns.

2. Purpose of the Project

The purpose of this project is to perform Market Basket Analysis on a transactional dataset to uncover patterns in customer purchasing behavior. Market Basket Analysis helps businesses gain deeper insights into which products are frequently bought together, enabling them to optimize their product placement, marketing strategies, and promotional campaigns. Here's why this project is significant and the business questions it aims to address:

Why Perform Market Basket Analysis on This Dataset?

Understanding Customer Behavior:

The dataset contains valuable information about customer purchases, such as item names, quantities, prices, and customer IDs. Analyzing this data helps identify purchasing patterns and trends that would otherwise remain hidden.

Optimizing Business Operations:

Retailers can use the insights gained to rearrange products in stores, design attractive offers, or suggest related products to customers during their shopping journey. This not only improves the shopping experience but also drives sales and increases revenue.

Addressing Data Science Objectives:

Market Basket Analysis serves as a practical example of applying data science techniques to solve real-world problems. It demonstrates the utility of algorithms like Apriori for deriving association rules, which businesses can directly apply.

Business Questions and Challenges Addressed

Which Products Are Frequently Bought Together?

This is the primary question Market Basket Analysis aims to answer. Knowing which products customers often purchase together can help businesses design more effective cross-selling strategies. For example, if bread and butter are commonly bought together, placing them close to each other or bundling them into a discount offer could boost sales.

What Are the Most Popular Items?

Identifying top-selling products enables businesses to manage inventory efficiently, ensuring that high-demand items are always in stock. This reduces the risk of missed sales opportunities.

How Can Product Placement Be Improved?

Insights from the analysis help retailers optimize the store layout. Products that are frequently bought together can be placed near each other, improving convenience for customers and increasing the likelihood of impulse purchases.

What Are the Opportunities for Marketing Campaigns?

With insights into purchasing patterns, businesses can create personalized marketing campaigns or promotional offers. For instance, discounts on items that are often bought with others (e.g., "Buy bread and get butter at 20% off") can encourage customers to buy more.

How Can Customer Segmentation Be Enhanced?

Analyzing customer behavior from the dataset provides insights into spending habits and preferences. This can help businesses segment customers based on their purchasing patterns and target them with tailored offers.

How Can Profitability Be Increased?

By understanding the relationships between products and customer preferences, businesses can focus on promoting items with high profitability margins that are frequently purchased with other products.

3. Dataset Overview

```
data = pd.read_csv("market_basket_dataset.csv")
print(data.head())
```

	BillNo	Itemname	Quantity	Price	CustomerID
0	1000	Apples	5	8.30	52299
1	1000	Butter	4	6.06	11752
2	1000	Eggs	4	2.66	16415
3	1000	Potatoes	4	8.10	22889
4	1004	Oranges	2	7.26	52255

Describe the Dataset Structure

This section provides a detailed overview of the dataset used for Market Basket Analysis. The dataset contains transactional data that includes the following columns:

1. BillNo: A unique identifier for each transaction or invoice.
2. Itemname: The name of the items purchased in each transaction.
3. Quantity: The number of units purchased for each item in a transaction.
4. Price: The cost of each item in the transaction.
5. CustomerID: A unique identifier for each customer who made the purchase.

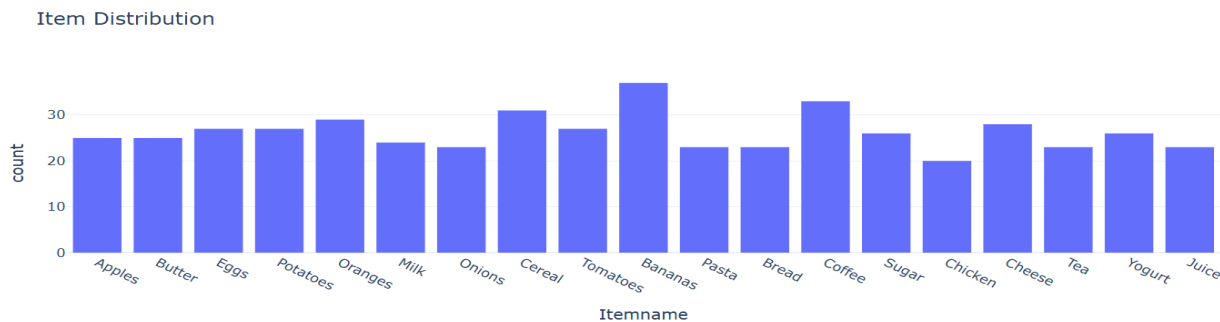
Each column is associated with specific data types, such as numbers for Quantity, Price, and CustomerID, and text for Itemname. This structure helps us analyze the data to identify buying patterns.

Highlight Key Statistics

This section summarizes the important statistics about the dataset:

- Number of Transactions: The dataset contains 500 unique transactions, identified by the BillNo column.
- Unique Items: There are several unique items sold in the store, such as Apples, Butter, Eggs, and Potatoes.
- Number of Customers: The dataset includes 500 customers, each with a unique CustomerID.
- Summary Statistics:
 - Average quantity of items purchased per transaction: 2.98.
 - Average price of items: \$5.62.
 - Minimum and maximum values for price and quantity give insights into the range of transactions.

4. Analysis



Sales Distribution of Items

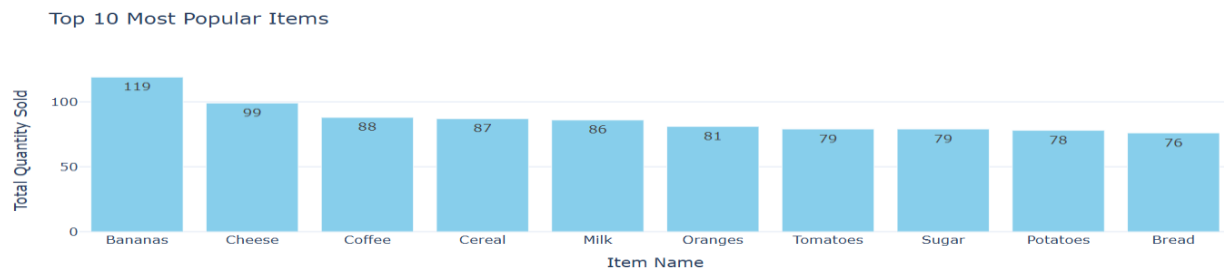
The first analysis focuses on the distribution of items sold by the store. A histogram was created to show how frequently each item appears in the dataset. This visualization helps to understand the overall sales pattern for different items.

Code Explanation:

`px.histogram(data, x='Itemname', title='Item Distribution')`: This line creates a histogram that plots the frequency of each item purchased. The `x='Itemname'` tells the program to consider the item names on the x-axis, and the title explains the chart's purpose.

`fig.show()`: This displays the histogram.

The chart helps identify which items are sold more frequently, providing insights into popular products in the store.



Top 10 Most Popular Items Sold

Next, we calculate the top 10 most popular items based on the total quantity sold. We group the data by the item name and sum up the quantities sold for each item. The top items are then displayed in a bar chart for easy comparison.

Code Explanation:

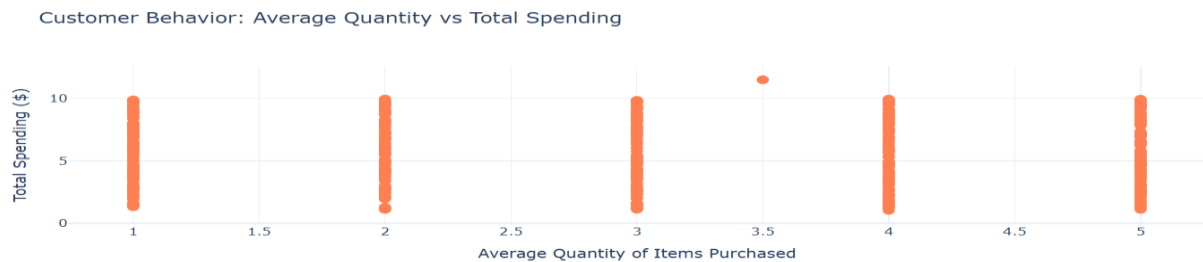
`item_popularity = data.groupby('Itemname')['Quantity'].sum().sort_values(ascending=False)`: This groups the dataset by Itemname and sums up the total quantity for each item. The `sort_values(ascending=False)` sorts the items from most sold to least sold.

`top_n = 10`: This defines how many top items we want to display (in this case, 10).

The `go.Figure()` and `fig.add_trace(go.Bar(...))` lines create a bar chart to visualize the popularity of the top 10 items.

`fig.show()`: This displays the chart.

From this analysis, we find that bananas are the most popular item sold in the store, which helps us understand consumer preferences.



Customer Behavior

	CustomerID	Average Quantity	Total Spending
Total Spending	10504	1	2.04
	10588	5	5.5
	10826	1	5.67
	11113	3	8.84
	11267	1	8.87
	11373	2	6.69
	11430	3	4.85
	11430	3	4.85
	11430	3	4.85
	11430	3	4.85
		Average Quantity	

Customer Behavior Analysis

The third part of the analysis explores customer behavior by calculating the average quantity of items purchased and total spending for each customer. This helps to identify how much each customer buys on average and how much they spend in total.

Code Explanation:

`customer_behavior = data.groupby('CustomerID').agg({'Quantity': 'mean', 'Price': 'sum'}).reset_index():` This line groups the data by CustomerID and calculates:

mean for Quantity: the average number of items purchased by each customer.

sum for Price: the total amount spent by each customer.

The `table_data` DataFrame is created to organize this information for display in a table.

`go.Figure()` creates a subplot that includes both a scatter plot and a table:

The scatter plot plots each customer's average quantity of items against their total spending.

The table shows the exact values of each customer's ID, average quantity, and total spending.

fig.show(): This displays the figure, showing both the scatter plot and table.

This analysis helps us understand how each customer behaves:

How much they buy on average (Average Quantity).

How much they spend in total (Total Spending).

Apriori algorithm

In this section, we apply the Apriori algorithm to generate association rules from the transactional data. The Apriori algorithm is widely used to discover frequent itemsets in large datasets, helping identify products that are often purchased together. By uncovering these patterns, businesses can make more informed decisions about product placement, promotions, and marketing strategies. Here's a step-by-step breakdown of the process:

1. Grouping Items by Transaction:

The first step is to group the data by BillNo (representing each transaction) and create a list of items bought in each transaction. This is done using the following line of code:

Code

```
basket = data.groupby('BillNo')['Itemname'].apply(list).reset_index()
```

This groups all the items bought in a single transaction together in a list format.

2. One-Hot Encoding:

To apply the Apriori algorithm, we need to encode the items as binary variables, where each item is represented as a column with a value of 1 if the item is present in the transaction and 0 if it is not. This is achieved by converting the item lists into a one-hot encoded dataframe using the following code:

Code

```
basket_encoded = basket['Itemname'].str.join('|').str.get_dummies('|')
```

This creates a binary matrix where each column corresponds to an individual item, and the values indicate whether an item is present in each transaction.

3. Finding Frequent Itemsets:

The Apriori algorithm is then applied to the encoded data to identify frequent itemsets—groups of items that appear together in transactions above a certain frequency threshold. The min_support parameter is used to set the minimum threshold for support, which represents the frequency of itemsets in the dataset. In this case, a min_support of 0.01 is chosen, meaning itemsets that appear in at least

1% of the transactions will be considered frequent. The algorithm is applied as follows:

Code

```
frequent_itemsets = apriori(basket_encoded, min_support=0.01, use_colnames=True)
```

4. Generating Association Rules:

Once the frequent itemsets are identified, we generate association rules. These rules indicate relationships between items, showing which items tend to be bought together. The lift metric is used to measure the strength of these associations. A min_threshold of 0.5 is applied to filter out weak associations. The rules are generated using the following code:

Code

```
rules = association_rules(frequent_itemsets, metric='lift', min_threshold=0.5)
```

The lift metric compares the actual frequency of item pairs being bought together against the expected frequency if they were independent, with higher values indicating stronger associations.

5. Displaying the Association Rules:

Finally, the top association rules are displayed, showing the antecedents (items on the left-hand side of the rule) and consequents (items on the right-hand side of the rule), along with their support, confidence, and lift values. These rules help identify which items are often bought together by customers, providing useful insights for decision-making.

5. Key Insights from the Market Basket Analysis:

	antecedents	consequents	support	confidence	lift
0	(Bread)	(Apples)	0.045752	0.304348	1.862609
1	(Apples)	(Bread)	0.045752	0.280000	1.862609
2	(Butter)	(Apples)	0.026144	0.160000	0.979200
3	(Apples)	(Butter)	0.026144	0.160000	0.979200
4	(Cereal)	(Apples)	0.019608	0.096774	0.592258
5	(Apples)	(Cereal)	0.019608	0.120000	0.592258
6	(Cheese)	(Apples)	0.039216	0.214286	1.311429
7	(Apples)	(Cheese)	0.039216	0.240000	1.311429
8	(Chicken)	(Apples)	0.032680	0.250000	1.530000
9	(Apples)	(Chicken)	0.032680	0.200000	1.530000

1. Frequently Purchased Together Items:

From the association rules, we observe certain items that are frequently bought together. For example:

- Bread and Apples: Customers who buy Bread often buy Apples as well (support of 4.58%). This indicates that these two items are commonly purchased together.
- Butter and Apples: Similarly, Butter and Apples also show a strong connection (support of 2.61%). This means customers tend to buy these items together.
- Cereal and Apples: There's also a notable relationship between Cereal and Apples (support of 1.96%).

2. Confidence:

The confidence values tell us the likelihood of buying one item when the other is already in the basket. For instance:

- There's a 30.43% chance that if a customer buys Bread, they will also buy Apples. This shows a moderate probability that customers buying one item are likely to purchase the other as well.
- Butter and Apples have a lower confidence value (16%), suggesting a slightly weaker but still meaningful connection between these two items.

3. Lift:

The lift values help us understand how much stronger the association between the items is, compared to how often we would expect them to be bought together by chance:

- A lift of 1.86 for Bread and Apples indicates that customers are 1.86 times more likely to purchase these items together than if they were independent of each other.
- Other items like Butter and Apples have a lift of 0.98, which is close to 1, indicating that the association is weaker compared to Bread and Apples.

6. Recommendations:

1. Product Placement and Promotions:

- Based on the frequent associations found, businesses can place Bread and Apples together in the store or offer bundle promotions. Since they are often bought together, such strategies could drive more sales by encouraging customers to purchase both items at once.

- Similarly, placing Butter and Apples near each other in the store could increase the likelihood of customers buying both items, especially if promotional offers or discounts are applied.

2. Targeted Marketing Campaigns:

- With the understanding of customer behavior patterns, businesses can create personalized marketing campaigns. For example, if a customer buys Bread, they could be offered a discount or coupon for Apples in future purchases, taking advantage of the 30.43% confidence.
- For customers who often purchase Cheese and Apples together, targeted ads or emails suggesting promotions for both could increase sales.

3. Cross-Selling Opportunities:

- Based on the lift values, items with higher lifts, like Bread and Apples, suggest stronger relationships. Businesses should focus on cross-selling these items together, whether through online shopping carts or in-store signage. For items with lower lifts, such as Butter and Apples, a more subtle approach may be needed, like suggesting them as complementary products.

7. Business Benefits:

1. Increased Sales:

By recognizing and leveraging these purchasing patterns, businesses can encourage customers to buy more items together, leading to an increase in average order value (AOV). Bundle offers or cross-selling promotions can boost revenue per transaction.

2. Improved Customer Engagement:

Personalized recommendations based on the analysis will make the shopping experience more relevant to customers. This can increase customer loyalty as they feel that the business understands their preferences and habits.

3. Efficient Inventory Management:

By identifying popular product combinations, businesses can better manage inventory and ensure they have enough stock of frequently bought together items. This can lead to fewer stockouts and more efficient restocking strategies.

8. Conclusion:

In this project, we applied Market Basket Analysis to a transactional dataset, uncovering frequent item associations and offering actionable recommendations for businesses to enhance their operations. The key insights from the analysis, such as the frequent

purchase of Bread and Apples, provide concrete opportunities for product bundling, cross-selling, and targeted promotions that can boost sales and improve the overall customer experience. This analysis not only demonstrates the practical application of data science in the retail industry but also highlights how businesses can leverage the power of data to optimize their strategies, maximize sales, and improve customer satisfaction.